Exp No:1.d

Conduct an experiment to encrypt and decrypt given sensitive data.

Aim:

To conduct an experiment to encrypt and decrypt given sensitive data.

Description:

Use the cryptography library to encrypt and decrypt a piece of data.

Code:

```
# Generate key and encrypt data
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
b'...'
f.decrypt(token)
b'Rajalakshmi Engineering College'
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)
# Decrypt data
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

OUTPUT:

Original Data: b'Rajalakshmi Engineering College.'

Encrypted Data: b'gAAAAABmmnHEhy1ZC1-Dox-URNAvYqFFGfJemMb_fKmXqhATe3AAn

XEOnSzaNRINI4zAbNK8lgx2pprrNn-j6agbWDf30OwbrZsBuz8fnDdX7N35WwloDu3m6pVv

kwv8GYDSI8Fov1tV'

Decrypted Data: b'Rajalakshmi Engineering College.'

Result :

Thus an experiment to encrypt and decrypt given sensitive data is conducted.